



Objective

As a Master's student in Robotics at University of Pennsylvania, I'm excited to leverage my software engineering and robotics expertise in an innovative project or research endeavor, where I can contribute to the development of cutting-edge technologies and collaborate with a diverse team.

Education

(Expected Graduation Date May 2027) University of Pennsylvania

- Masters of Science and Engineering in Robotics (MSE ROBO)
- **Current Classes:** ESE 6500: Learning in Robotics, ESE 6510: Physical Intelligence, CIS 5800: Machine Perception
- **Previous Classes:** MEAM 5170: Control and Optimization in Robotics, MEAM 5200: Introduction to Robotics, ENM 5500: Mathematics in Robotics

University of California at Irvine

- Bachelor's of Science in Computer Science and Engineering Major (CSE)
- UC GPA of 3.83 / 4.0, Graduated June 2023

Work / Research Experience

JIRL Research / DARPA Challenge w/ Antonio Loquercio (UPenn Robotic Research) (Oct 2025 - Feb 2025)

- Develop a semantic search framework (AI / LLMs) for autonomous robots, building upon [Vision-Language Frontier Maps \(VLFM\)](#) to enable advanced capabilities for a DARPA competition
- Paper Co-author: **RAVEN: A General Visual Memory for Long-Horizon Robot Reasoning and Navigation** - Under review at Robotics: Science and Systems (RSS), developed tooling for experiment and evaluation workflow
- Utilize tools such as ROS, IsaacSim, Gazebo, Ollama / LLM Models, Tensor, Numpy, etc.

Amazon SDE (Jan 2024-May 2025 Full Time, 2022 Summer Internship)

- Work on Amazon Photos Sharing and Engagement Team
- Design and write Back-End Code that's deployed into Production and have direct impact on Customers
- Utilize AWS Tools/Services including DynamoDB, Gluejob, Athena, CDK, etc. to analyze customer data and provide significant information to the user to maximize delight

Quadruped RL Researcher w/ Professor Khargonekar (UCI Robotics Research) (Aug 2023-Jan 2024)

- Conduct research extending Jitendra Malik's paper on Rapid Motor Adaptation by employing a Graph Neural Network in junction with any quadrupeds URDF to create a universal model
- Utilize Weight and Biases to conduct a series of experiments to dictate the performance of our models
- Add Domain Randomization and Rewards to Isaac Gym

Chief Engineer at UAV Forge (Drone Team) (2021-2023)

- Manage the Electrical and Software Engineering teams, including structuring, hiring, job assignment, team road map, and other managerial roles
- Implement them with Autonomous Flying Vehicles and Autonomous Driving Vehicles
- Interface between Robots and ground station with ROS (Robot Operating System) and Linux

Researcher at Professors Kia's Robotics Lab (UCI Robotics Research) (2022-2023)

- Perform research and experiment on local and global motion planning algorithms
- Model control systems (LQR, MPC) in regards to various kinematic / dynamic systems
- Utilize Python, Matplotlib, Numpy, and Drake to simulate and implement control systems

Skills, Abilities and Honors

- C#, Python, C++, Java, Javascript, Typescript, HTML, CSS, Git, Docker, Arduino, PlatformIO, Unity, AWS, Typescript, ROS, Linux; Experienced in Python Libraries Numpy, Matplotlib, Jupyter Notebook, and Scipy, used in Machine Learning and Vision, Isaac Gym / Isaac Sim for Robot Simulation and Training, Weights and Biases
- **Fluent in English and Proficient in Spanish**